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$$\begin{aligned}\lim_{x \rightarrow \infty} \left(\frac{x+1}{x-1} \right)^{-x} &= \lim_{x \rightarrow \infty} \left(\frac{x-1+1+1}{x-1} \right)^{-x} \\&= \lim_{x \rightarrow \infty} \left(\frac{x-1}{x-1} + \frac{2}{x-1} \right)^{-x} = \lim_{x \rightarrow \infty} \left(1 + \frac{2}{x-1} \right)^{-x} \\&= \lim_{x \rightarrow \infty} \left(1 + \frac{2}{x-1} \right)^{\frac{x-1}{2} \cdot \frac{2}{x-1} \cdot (-x)} \\&= \lim_{x \rightarrow \infty} \left[\left(1 + \frac{2}{x-1} \right)^{\frac{x-1}{2}} \right]^{\frac{-2x}{x-1}} \\&= e^{\lim_{x \rightarrow \infty} \left(\frac{-2x}{x-1} \right)} = e^{\lim_{x \rightarrow \infty} \left(\frac{-2x}{x} \right)} \\&= e^{\lim_{x \rightarrow \infty} -2} = e^{-2} = \frac{1}{e^2}\end{aligned}$$

References